Predicting Braessian Edges in Oscillator Networks

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INTRODUCTION

- **Network Synchronization**
  - Many systems (power grids, traffic, neurons) need their components **to move together (synchronize).**
  - We model this using **the Kuramoto Model** to assess a **network of nodes.**
  - Whether the system synchronizes depends on the **coupling strength**.
- **The Problem**
  - Adding a new connection **(an edge)** should help a network synchronize, but sometimes it makes things worse.
  - The network then **needs stronger coupling** to stay aligned.
  - This counter-intuitive failure is **Braess's Paradox.**

Our Goal: Can we predict which added edges will cause Braess's paradox in oscillator networks?

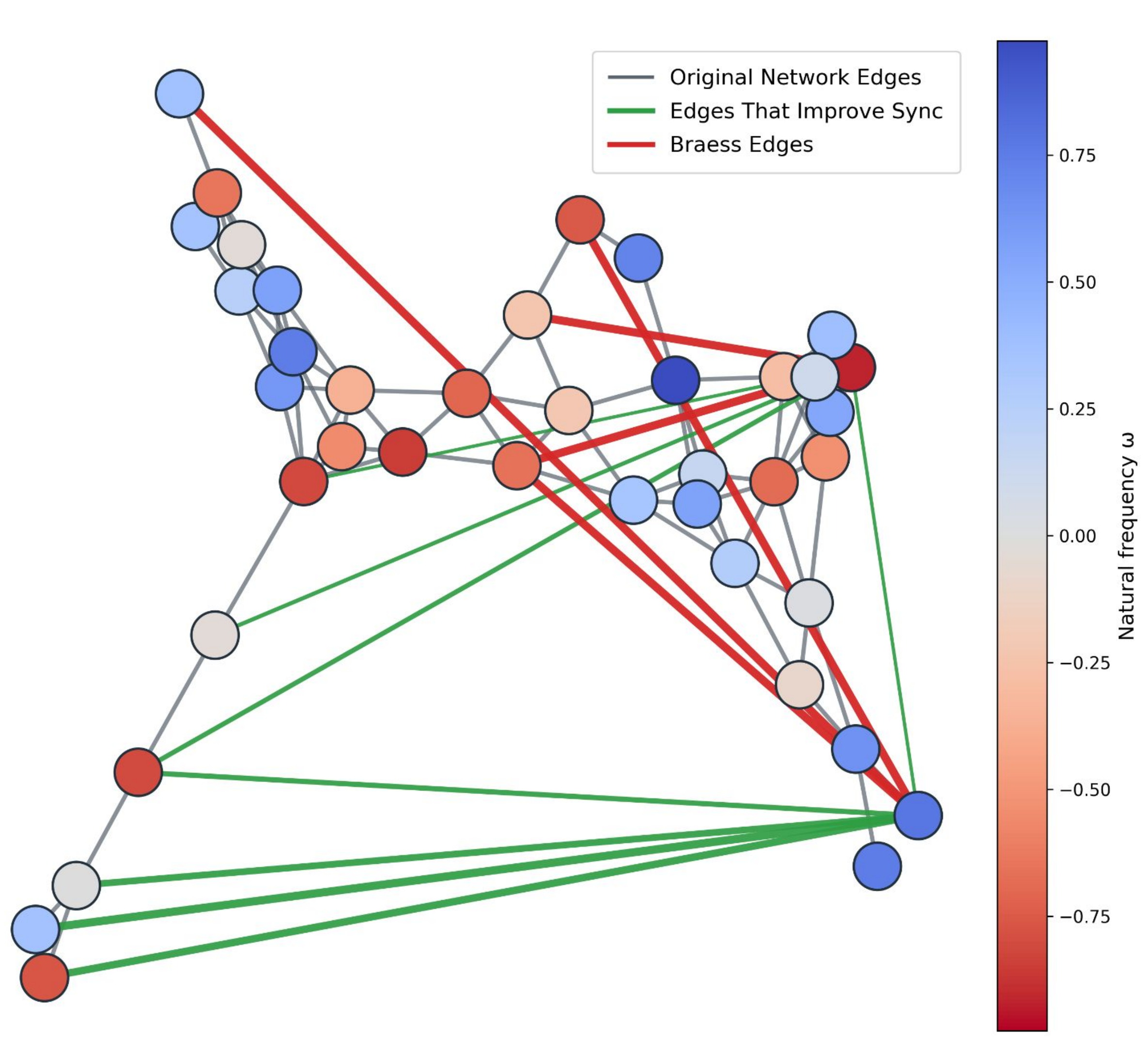
The Kuramoto Model

Phase of oscillator i | Coupling strength constant | Interaction between i and j pulling phases

$$\frac{d\theta_i}{dt} = \omega_i + K\sum_{j=1}^{N} W_{ij}\sin(\theta_j - \theta_i)$$

Natural frequency of oscillator i | Adjacency matrix | Total influence of all of i's neighboring nodes

Network Example



METHODOLOGY: DERIVING THE THEOREM

Setup

- Represent the new edge with a parameter a (its weight; a=0 no edge).
- **View** the critical coupling as a function $\mathbf{K_c(a)}$.
- Apply the Implicit Function Theorem (IFT) to **compute $K_c'(0)$**

Observations about the Kuramoto Model
- At K_c the Jacobian is **singular**
- IFT requires a full rank Jacobian

⚠ We **CANNOT** apply the IFT directly

Augmented System

$$G(\theta, v, K, a) = \begin{pmatrix} F(\theta, K, a) \\ D_\theta F(\theta, K, a)\, v \\ v^\top v - 1 \\ v^\top \mathbf{1} \\ \theta_1 \end{pmatrix} = 0$$

Each row of G enforces a condition:
- Steady state equation (when our system is in sync).
- Critical point where our system loses stability (eigenvalue of v = 0).
- Normalized eigenvector v to remove scaling freedom.
- Orthogonality for v to the trivial direction.
- Gauge fix to stop rotational symmetry

Our new Augmented System G maps

$$\mathbb{R}^{2n+2} \longrightarrow \mathbb{R}^{2n+3}$$

New Obstacle: Our system has more equations than unknowns, can't apply IFT.

To address that problem, we can apply the Rank–Nullity Theorem → We can take the derivative of G and solve for the co-kernel. →

$$\psi_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad \psi_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

- The cokernel of DG is two-dimensional, spanned by ψ_1 and ψ_2.
- Projection onto ψ_1 and ψ_2 removes those unsolvable direction.
- This restores the right dimensions so we can apply IFT.

$$\psi_i^T DG = 0.$$

$$P\mathbf{w} = \mathbf{w} - (\mathbf{w}^T\psi_1)\psi_1 - (\mathbf{w}^T\psi_2)\psi_2, \quad \mathbf{w} \in \mathbb{R}^{2n+3}$$

$$PG = H \longrightarrow H : \mathbb{R}^{2n+2} \to \mathbb{R}^{2n+1}$$

Final Derivative

$$K'(0) = \sin\big(\theta_p(0) - \theta_q(0)\big)\, K^2(0)\, \frac{v_q(0) - v_p(0)}{v^T\Omega}$$

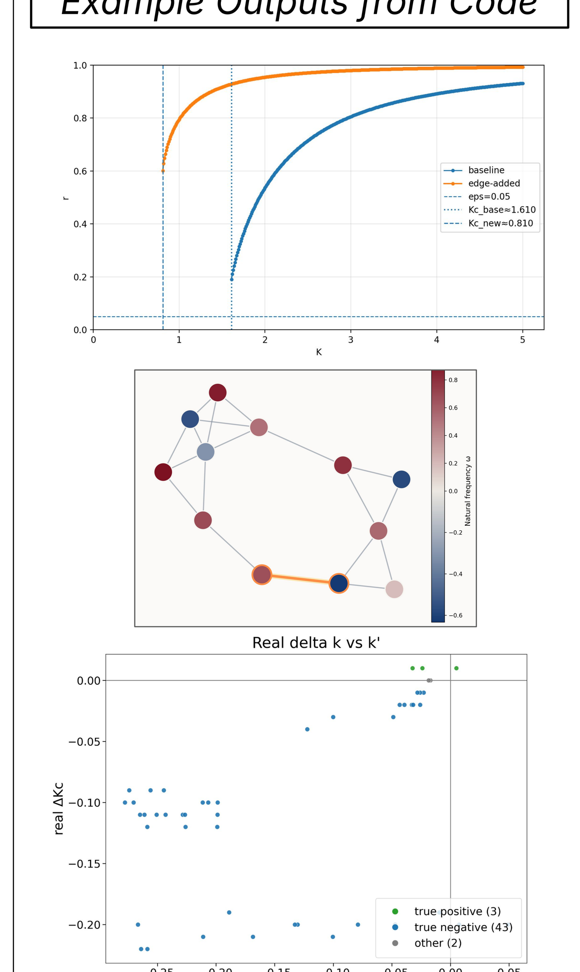
- $K'(0) < 0 \to K_c$ decreases
- $K'(0) = 0 \to K_c$ first-order change vanishes
- $K'(0) > 0 \to K_c$ increases → Braess Edge

COMPUTATIONAL METHODOLOGY & MOTIFS

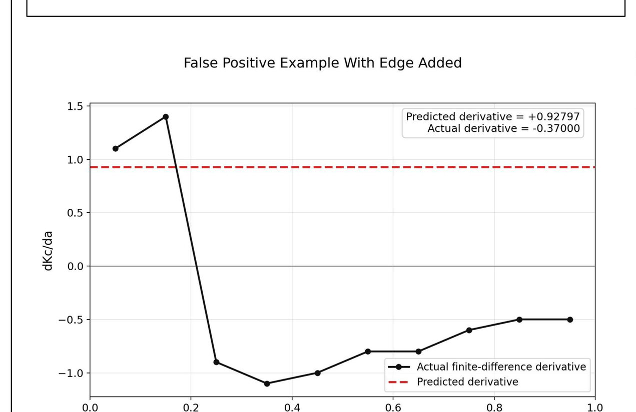
Dataset Generation

Generate (Random) Erdos-Reynl or Random Geometric Graphs
↓
Sweep from High K. Step down Δ K. Find K_c
↓
Add A Missing Edge
↓
Sweep down from High K, Step down Δ K. Find New K_c
↓
Save all values to dataset
↓
Run Prediction: Calculate K'(0)

Example Outputs from Code

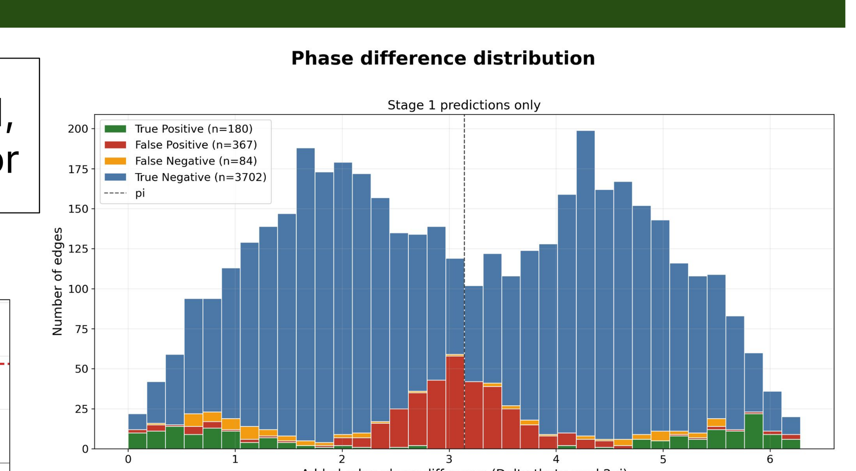


Motif/Discovery #1: As a → 1, k_c exhibits non-linear behavior

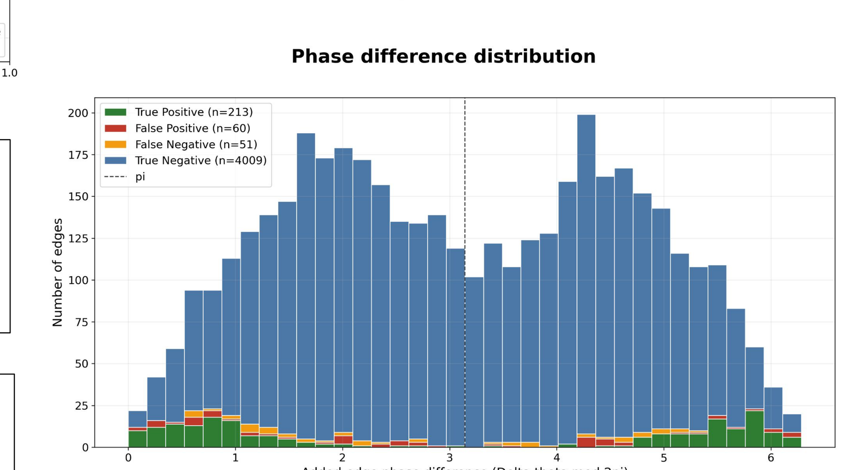


Motif/Discovery #2: When Oscillators cross threshold near π → sign changes in $k_c(a)$ graph as a → 1

Motif/Discovery #3: Added edge between nodes with similar eigenvector values → false negative

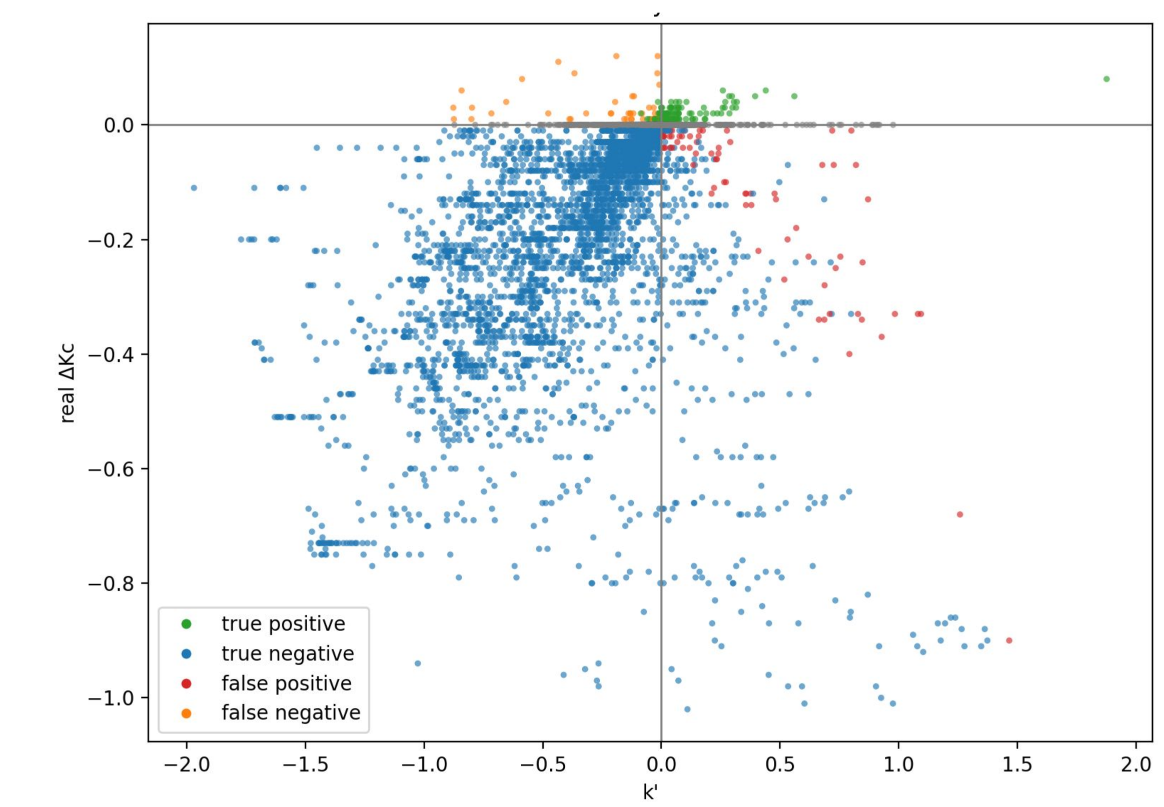


Without Motif Check Extensions



With Motif Check Extensions

RESULTS



Real ΔK vs Predicted ΔK for 10-14 Node Dataset

	Predicted Braess	Predicted Non-Braess
True Braess	1,944	152
True Non-Braess	309	29,742

Only edges with $|\Delta K_c| \geq 0.01$ were included in evaluation.

Most edges aren't Braess, but those that are can signal counterintuitive instability that **risks** harmful outcomes (e.g., in power grids).

Dataset	# Networks	#Edges	*Accuracy*
10-14 Nodes	200	9,662	99.1%
20-28 Nodes	196	40,825	98.6%

DISCUSSION & CONCLUSION

- We **derived** and **proved** an exact **theorem** for the effect of edge addition on critical coupling
  - Using IFT and projection-based techniques, we obtained an **exact formula** for $\mathbf{K_c'(0)}$
  - This provides a rigorous local predictor at a = 0
- The full transition from a = 0 to a = 1 reveals nonlinear behavior beyond first-order theory
  - Geometric frustration and edges near π phase difference explain many paradoxical cases
  - Changes in oscillator ordering + near π help detect when the local predictor fails globally
- These motifs turn the local theorem into a **highly accurate** global prediction framework
  - They explain many discrepancies between local and final behavior
  - Incorporating them raises prediction **accuracy** to about **99%**

Future Work

- Continue refining motifs to find the final cases were local prediction doesn't match global behavior.
- Apply the classifier to directed Kuramoto networks.

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